

Multiple Choice (10 marks)

1. If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel?
 - (a) 120 miles
 - (b) 80 miles
 - (c) 26 miles
 - (d) 12 miles
2. Stephanie flew 2,448 miles from Los Angeles to New York City. What is the number of miles Stephanie flew rounded to the nearest thousand?
 - (a) 2,000
 - (b) 2,400
 - (c) 2,500
 - (d) 3,000
3. Write an equation for the line tangent to the curve defined by $F(t) = (t^2 + 1, 2^t)$ at the point where $t = 2$.
 - (a) $y - 4 = \ln 2(x - 2)$
 - (b) $y - 4 = 4 \ln 2(x - 2)$
 - (c) $y - 4 = 4(x - 5)$
 - (d) $y - 4 = \ln 2(x - 5)$
4. What is the shortest distance from the origin to the circle defined by $x^2 - 24x + y^2 + 10y + 160 = 0$?
 - (a) 10
 - (b) 16
 - (c) 24
 - (d) 12
5. Keri spent 3 hours on her homework. She spent equal amounts of time on all subjects. If Keri spent 1 over 2 hour on science, how many subjects did she study?
 - (a) 4 subjects
 - (b) 2 subjects
 - (c) 6 subjects
 - (d) 5 subjects

True / False (10 marks)

Answer True or False

6. True or False: There are 45 even positive integers less than the product of 7 and 13.
7. True or False: Forty-nine students on the dance team take tap lessons, which is more than half of 100.
8. True or False: The ratio of the number of pears to the number of apples in Simon's fruit salad is 1 to 2.
9. True or False: The solution to the equation $14 = w + 23$ is $w = -9$.
10. True or False: The sum of the three real numbers not in the domain of the function is 0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that can influence the value of the correlation coefficient r , and explain why r is not a measure of causation.
12. Analyze the set $\{2^{\frac{n}{o}} \mid n \text{ is a positive integer, } o \text{ is a positive integer, and } n \text{ is not divisible by } o\}$.

End of Paper